Clinical Reasoning & Explainability Report

Patient: Ignacio Dach | Cycle #30

1. Patient Context

Cancer Type: LUNG
Age: 88 | Gender: M
Chemotherapy Drugs: Cisplatin, Paclitaxel

2. Clinical Data Inputs & Their Impact

2a. Laboratory Values

Lab Test	Value	Normal	Status	Impact on Side Effects
WBC	2.6 x1000/mm3	4.5-11.0	LOW	Increased fever/infection risk and fatigue severity
Hemoglobin	16.5 g/dL	12-18	NORMAL	No modifier applied
GFR	12.0 mL/min	>60	LOW	Increased nausea/fatigue (drug toxicity accumulation)
Pain Score	9/10	0-3	HIGH	Increased anxiety/depression and muscle/joint pain

2b. Wearable Biometric Data

Metric	Value	Normal	Status	Impact
HRV (RMSSD)	27.6 ms	>30 ms	LOW	Autonomic stress -> increased fatigue, anxiety, SOB
Sleep Efficiency	92%	>85%	NORMAL	Adequate sleep quality
Mean Heart Rate	74 bpm	60-100	INFO	Contextual cardiovascular data

2c. Age Modifier

Age 88: 20% severity increase applied (elderly patients have reduced drug clearance)

3. Drug Side-Effect Profiles

Base severity weights for each drug (0-1 scale, higher = more likely/severe)

Cisplatin -- Known Side Effect Affinities

Nausea & Vomiting	85%
Fatigue	70%
Loss of Appetite	65%
Anxiety / Depression	40%
Numbness / Tingling	30%
Diarrhea	30%
Muscle / Joint Pain	30%
Fever	25%
Sore Mouth	20%
Shortness of Breath	20%
Constipation	20%
Edema (Hands/Feet)	15%
Itching / Rash	10%
Swallowing Difficulty	10%

Paclitaxel -- Known Side Effect Affinities

Numbness / Tingling	80%
Muscle / Joint Pain	75%
Fatigue	70%
Nausea & Vomiting	50%
Loss of Appetite	45%
Diarrhea	35%
Anxiety / Depression	35%
Sore Mouth	30%
Edema (Hands/Feet)	30%
Fever	25%
Constipation	25%
Shortness of Breath	20%
Itching / Rash	20%
Swallowing Difficulty	10%

4. Temporal Severity Model

Side-effect severity follows a temporal curve across the 7-day chemo cycle. Day 1 (infusion) starts mild, severity peaks at Days 3-5 (nadir period when blood counts are lowest), then eases on Days 6-7 (recovery). This pattern is well-established in oncology literature.

Day	Multiplier	Phase
Day 1	0.4x	Infusion
Day 2	0.7x	Onset
Day 3	1.0x	Nadir (Peak)
Day 4	1.0x	Nadir (Peak)
Day 5	0.9x	Late Nadir
Day 6	0.6x	Recovery
Day 7	0.4x	Recovery

5. Per-Side-Effect Severity Reasoning

Final Grade = Drug Base x Clinical Modifier x Temporal Curve x Cycle Factor + Random Noise

Fever	Mild	Mild	Mode	Mode	Seve	Mild	Mild	Cisplatin base affinity: 25% | Paclitaxel base affinity: 25% | Low WBC (2.6) amplifies this effect
Fatigue	Seve	Seve	Seve	Seve	Seve	Seve	Seve	Cisplatin base affinity: 70% | Paclitaxel base affinity: 70% | Low WBC (2.6) amplifies this effect | Low HRV (27.6ms) indicates autono
Nausea & Vomiting	Seve	Seve	Seve	Seve	Seve	Seve	Seve	Cisplatin base affinity: 85% | Paclitaxel base affinity: 50%
Sore Mouth	Mild	Mode	Mode	Mode	Mode	Mild	Mild	Cisplatin base affinity: 20% | Paclitaxel base affinity: 30%
Diarrhea	Mild	Mild	Mode	Seve	Mode	Mild	Mild	Cisplatin base affinity: 30% | Paclitaxel base affinity: 35%
Constipation	Mild	Mild	Mild	Mode	Mild	Mild	None	Cisplatin base affinity: 20% | Paclitaxel base affinity: 25%
Loss of Appetite	Mode	Seve	Seve	Seve	Seve	Mode	Mild	Cisplatin base affinity: 65% | Paclitaxel base affinity: 45%
Swallowing Difficulty	None	None	None	Mild	None	None	None	Cisplatin base affinity: 10% | Paclitaxel base affinity: 10%
Anxiety / Depression	Mild	Seve	Seve	Seve	Seve	Mode	Mild	Cisplatin base affinity: 40% | Paclitaxel base affinity: 35% | Low HRV (27.6ms) indicates autonomic stress
Edema (Hands/Feet)	Mild	Mild	Mild	Mode	Mode	Mode	Mild	Cisplatin base affinity: 15% | Paclitaxel base affinity: 30%
Itching / Rash	None	Mild	Mild	Mild	Mild	Mild	None	Cisplatin base affinity: 10% | Paclitaxel base affinity: 20%
Shortness of Breath	None	None	Mild	Mild	Mild	None	None	Cisplatin base affinity: 20% | Paclitaxel base affinity: 20% | Low HRV (27.6ms) indicates autonomic stress
Muscle / Joint Pain	Mode	Seve	Seve	Seve	Seve	Seve	Seve	Cisplatin base affinity: 30% | Paclitaxel base affinity: 75%
Numbness / Tingling	Mild	Seve	Seve	Seve	Seve	Seve	Mode	Cisplatin base affinity: 30% | Paclitaxel base affinity: 80%

6. Methodology & Data Sources

-- SEVERITY CALCULATION: Final severity = Drug_Base_Weight x Clinical_Modifier x Temporal_Curve x Cycle_Factor + Gaussian_Noise(0, 0.08)

-- DRUG PROFILES: Based on published chemotherapy side-effect incidence rates from oncology literature.

-- CLINICAL MODIFIERS: Lab values (WBC, Hemoglobin, GFR, Pain) adjust severity multipliers based on clinical thresholds.

-- WEARABLE DATA: HRV (RMSSD) and sleep efficiency provide real-time physiological stress indicators.

-- TEMPORAL MODEL: 7-day cycle follows established nadir pattern (peak toxicity Days 3-5 post-infusion).

-- CYCLE FACTOR: Later cycles accumulate toxicity at +2% per cycle (capped at 40% increase).

-- GRADING: Continuous 0-1 score mapped to 4 grades: None (<0.20), Mild (0.20-0.44), Moderate (0.45-0.69), Severe (>=0.70).

-- DATA SOURCES: Synthea synthetic patient data (demographics, conditions, medications, observations, encounters) + generated wearable biometrics.